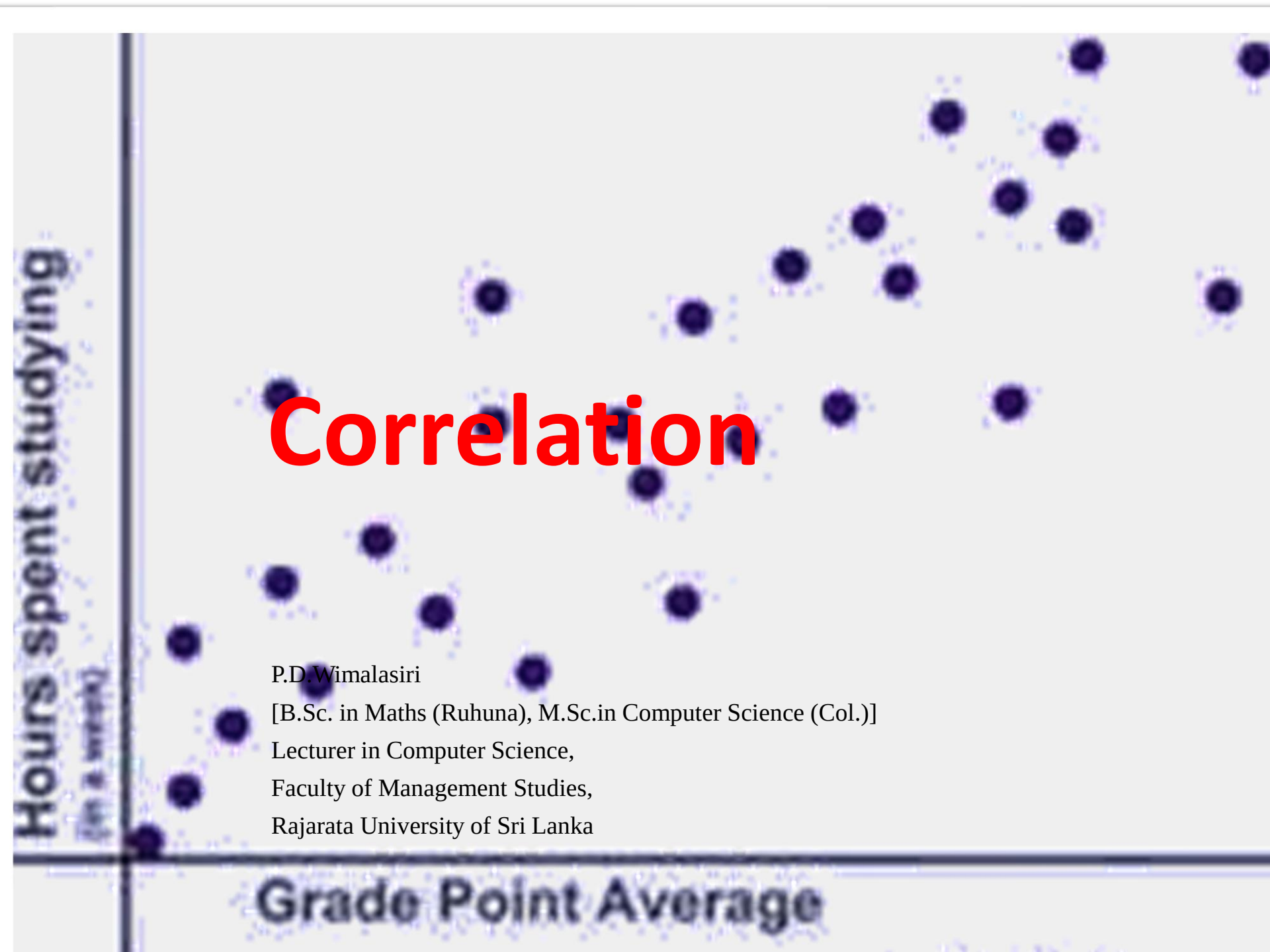




A scatter plot with 'Hours spent studying (per 2 weeks)' on the y-axis and 'Grade Point Average' on the x-axis. The plot shows approximately 20 data points, each represented by a solid black dot surrounded by a dashed circle. The points generally trend upwards from left to right, indicating a positive correlation between the two variables.

Correlation

P.D.Wimalasiri

[B.Sc. in Maths (Ruhuna), M.Sc.in Computer Science (Col.)]

Lecturer in Computer Science,

Faculty of Management Studies,

Rajarata University of Sri Lanka

Correlation measures the association between two variables.

Examples:

Quantity demand and price of a product.

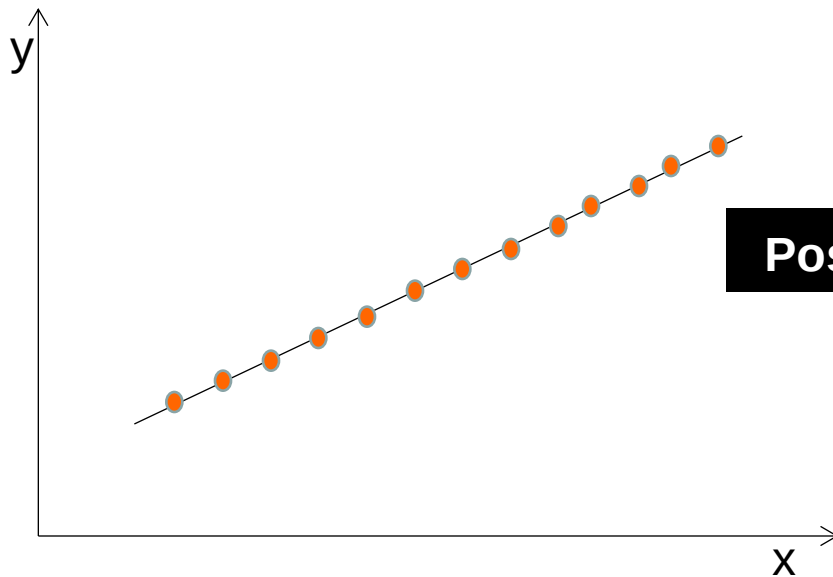
Sales amount and advertizing cost.

Marks obtained by a student for a subject and the time he spent on studing that subject.

Daily Sales amount of Ice cream and day temperature.

If two variable are plotted in a same graph it can be one of the following cases. These graphs are called **Scatter graphs**.

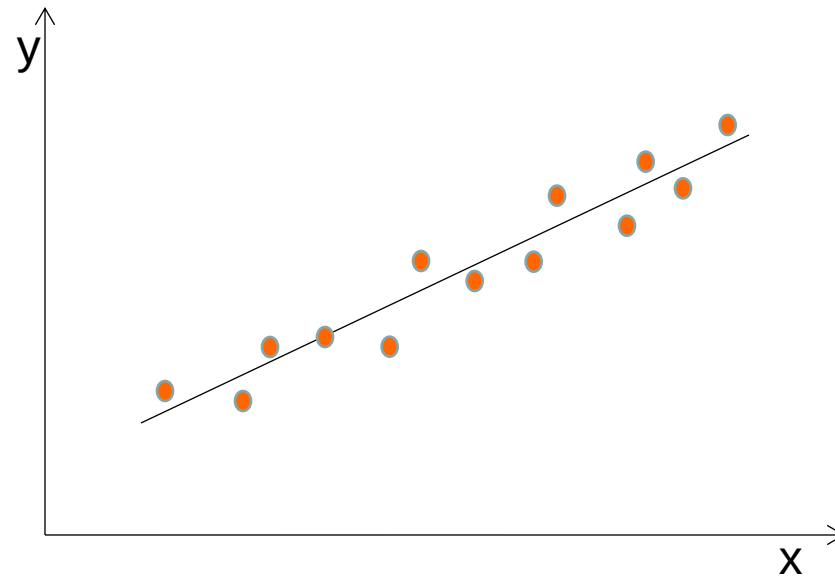
1. All the points are **on** a straight line.
When one variable increases other one also increases.



Positive Perfect correlation. $r = +1$

2. All the points are not on a straight line but most of them are **closer to the line**.

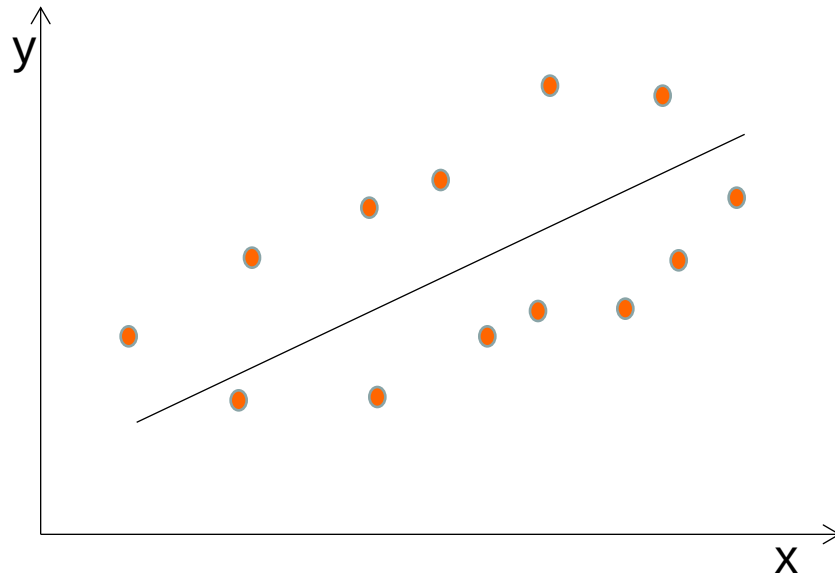
When one variable increases other one also increases.



Positive strong correlation.. $r \approx 1$
(r has a value closer to +1)

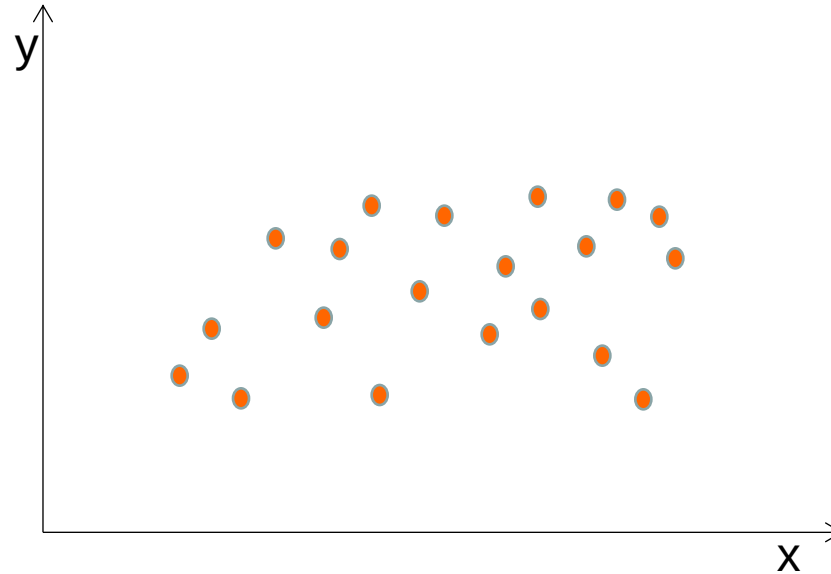
3. All the points are not on a straight line but most of them are **far away from the line** .

When one variable increases other one also increases.



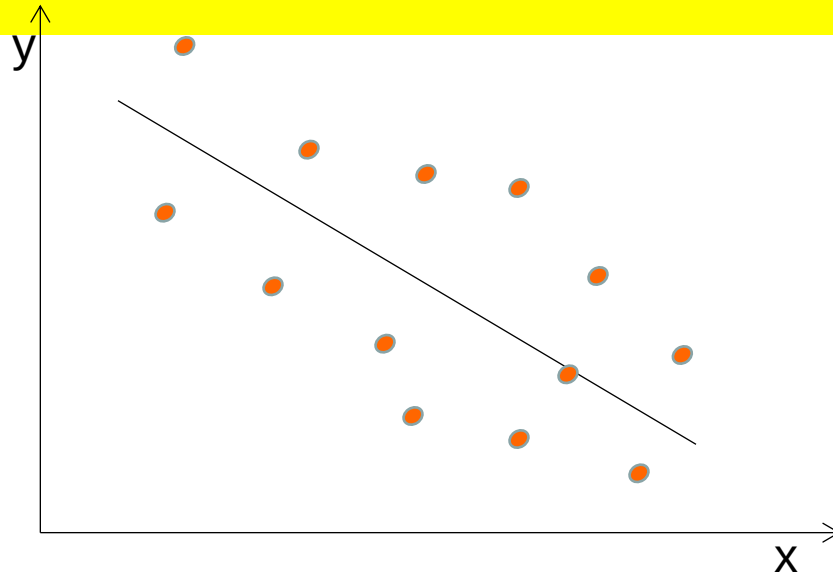
Positive weak correlation.. $r \approx 0$
(r has a value closer to 0)

4. All the points are randomly spread.



No correlation. $r=0$

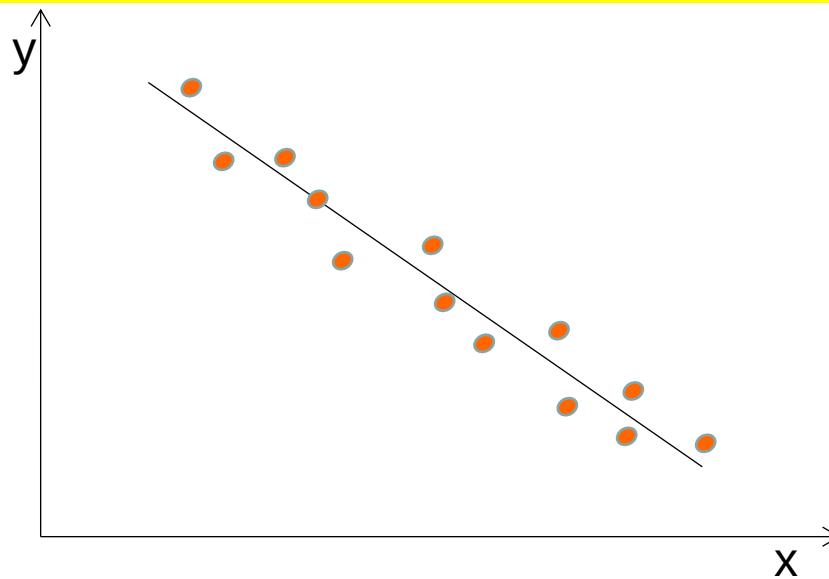
5 All the points are not on a straight line but most of them are **far away from the line** .
When one variable **increases** other one **decreases**.



Negativ eweak correlation.. $r \approx 0$
(r has a value closer to 0)

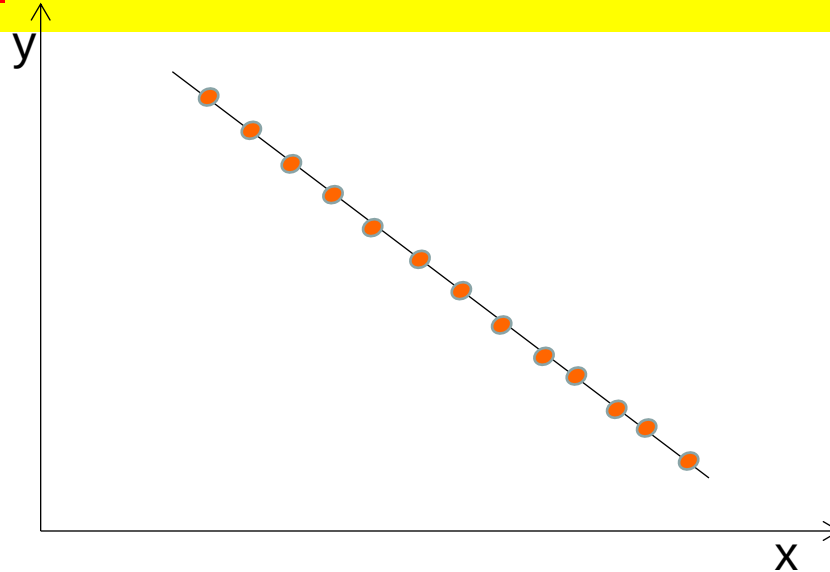
6. All the points are not on a straight line but most of them are **closer to the line** .

When one variable **increases** other one **decreases**.



Negative strong correlation.. $r \approx -1$
(r has a value closer to -1)

7. All the points are **on** a straight line.
When one variable increases other one decreases.



Negative Perfect correlation. $r = -1$

Calculation of coefficient of correlation.

Also known as Karl Pearson's coefficient of correlation.

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Calculation of coefficient of correlation.....

$$r = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\sqrt{\left\{\frac{\sum x^2}{n} - (\bar{x})^2\right\}\left\{\frac{\sum y^2}{n} - (\bar{y})^2\right\}}}$$

$$r = \frac{\sum (x - \bar{x})(y - \bar{y}) / N}{\sigma_x \sigma_y}$$

Example

x	y
10	24
20	38
30	55
40	88
50	109

Find the coefficient of correlation between var x and y given as above .

solution

x	y	xy	x ²	y ²
10	24	240	100	576
20	38	760	400	1444
30	55	1650	900	3025
40	88	3520	1600	7744
50	109	5450	2500	11881
150	314	11620	5500	24670

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

$$r = \frac{(5)(11620) - (150)(314)}{\sqrt{[(5)(5500) - (150)^2][(5)(24670 - (314)^2)]}}$$

$$r = \underline{\underline{0.989}}$$